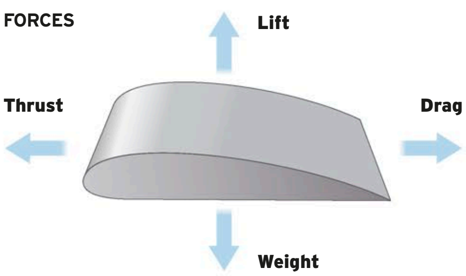
How Aircraft Fly

One thing that has not changed since the first heavier-than-air aircraft flew is the way wings work: a wing-based craft stays aloft because of a difference in air pressure of a few pounds or kilograms above and below the wings. The difference in pressure that allowed Sir George Cayley's coachman to glide across Brompton Dale in 1853 is the same pressure difference that allows supersonic aircraft to stay in the sky today. Even "stick-and-rudder" flight controls—originally devised by Louis Blériot for his monoplane crossing of the Channel in 1909—are found in today's aircraft, from ultralights to spaceplanes. Huge advances have been made in reducing aerodynamic drag and coping with the effects of supersonic airflow, but the fundamentals of wing-borne flight remain the same. Lift and balancing forces (weight, thrust, and drag) work together to generate flight. In level flight, lift balances the weight of the aircraft, and engine thrust balances drag.

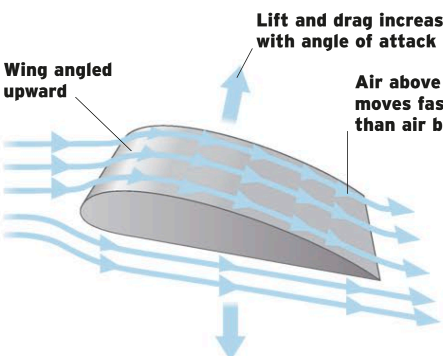
LIFT AND BALANCING FORCES

As a wing passes through air, the air moving across the top moves faster than the air passing beneath. This creates higher air pressure below the wing, which generates lift. Varying the angle of attack, at which the wing meets the oncoming air, affects the amount of lift and drag generated, and stalling, the loss of lift, occurs when the critical angle is exceeded.

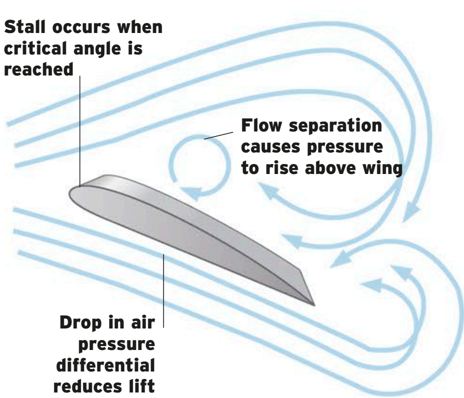
FORCES



MEDIUM ANGLE OF ATTACK



HIGH ANGLE OF ATTACK

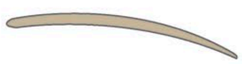


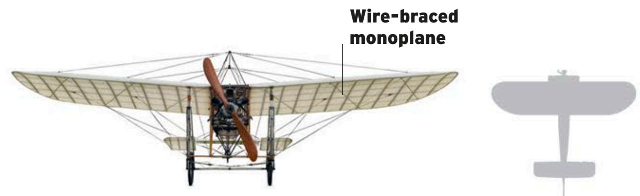
EVOLUTION OF WING SHAPES

Because the pioneering aviators were primarily concerned with building the lightest wings possible, they used very thin surfaces braced by wires. Thicker, less drag-inducing airfoil sections were used for World War I biplanes, but, until structural design improved at the time of the Spitfire, unbraced cantilever wings had to be made thicker for high-speed flight. Sweptback "laminar flow" wings ultimately proved superior for transonic flight, improved materials allowing them to be made thinner for even higher speed.

BLÉRIOT XI

The Blériot XI's thin, highly cambered (curved) airfoil was typical of its time. The shallow wings could be warped (twisted) for roll control, while bracing wires supported the bending load, kept to a minimum by the short span.





ROYAL AIRCRAFT FACTORY S.E.5A

By 1916, airplanes were routinely exceeding 100 mph (161 km/h) and performing aerobatic maneuvers. Flatter and low-drag airfoils could be kept relatively thin because of the inherent strength of the wire-braced biplane structure.

SPITFIRE

R. J. Mitchell's successful Schneider Trophy seaplanes had thin wings supported by bracing wires. Combining very strong, shallow spars with a skin that carried the load resulted in a cantilever wing that was thin and strong.

F-86 SABRE

North American's piston-engine P-51 was faster than the Spitfire due to its "laminar flow" wing, which was profiled to minimize drag-inducing turbulent flow. This swept laminar flow wing proved ideal for transonic flight.

LOCKHEED F-117 NIGHTHAWK

The very thin, sharp edged, and "over-swept" wing of the F-117 creates a minimal radar signature. Computer fly-by-wire controls compensate for instability and other aerodynamic deficiencies.